

Activity-1

Q) Develop a **Bank Account Management System** that performs deposit and withdrawal operations.

- Create a custom exception `InsufficientBalanceException`.
- Throw the exception when a withdrawal amount exceeds the available balance.
- Use `try-catch-finally` blocks to handle exceptions gracefully.
- Use the `throws` keyword to propagate exceptions from methods.
- Ensure that a transaction summary is printed in the `finally` block.

Q) Implement the **Producer–Consumer problem** using Java threads.

- Use a shared buffer with a fixed size.
- Implement synchronization using `wait()`, `notify()`, or `notifyAll()`.
- Ensure thread safety and avoid race conditions.
- Demonstrate correct inter-thread communication.